EcoAgent

Smart Energy Assistant

Estimate household energy consumption, electricity cost and associated emissions.

Project presentation | Output screenshots

Energy

kWh per appliance

Cost

₹ using state tariffs

Emissions

CO_2 at 0.708 kg/kWh

Why EcoAgent?

The problem



No appliance-wise visibility

Households can't tell which device drives the bill.



Only a monthly total

Meters give no link to cost in ₹ or CO_2.



Surprise bills

No guidance on which habits to change.

Our objective

- Estimate daily kWh for each appliance with a Random Forest model blended with the physics formula
- Convert usage into cost with state-wise tariffs and into CO_2 emissions
- Show monthly and yearly summaries, personalised saving tips and downloadable reports
- Give admins a view of all users and a state-wise yearly cost forecast

System architecture and tech stack



Streamlit UI

app.py

Login, usage form, tabs, admin dashboard



Prediction engine

tools.py

Random Forest + physics formula, cost, CO_2, live weather



SQLite database

db.py

users, usage and summary tables



Outputs

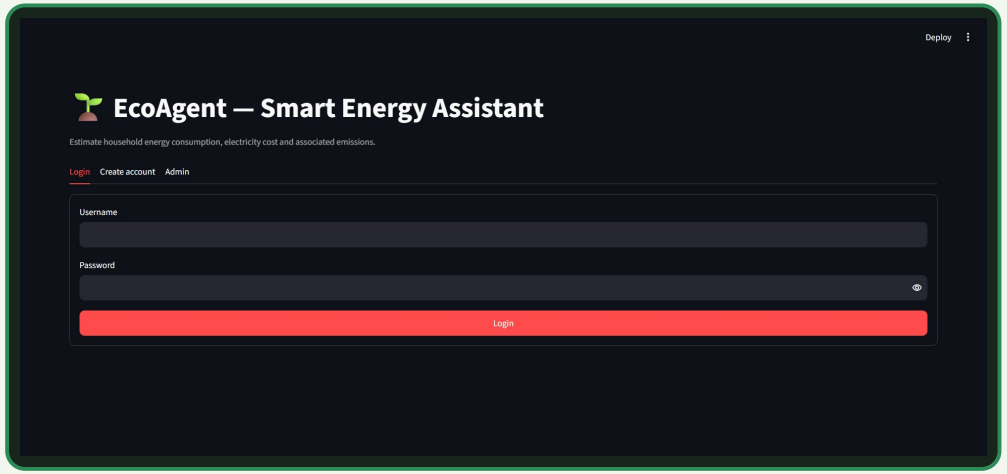
reports

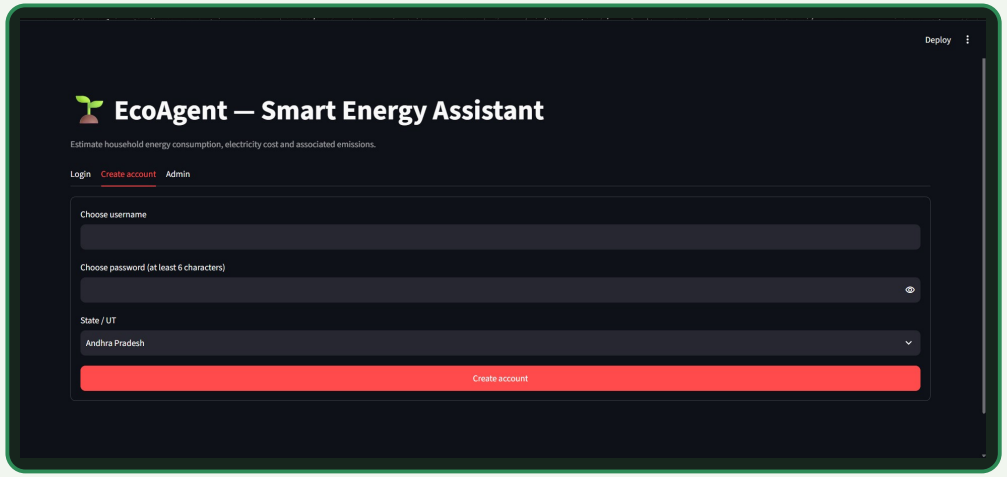
Summaries, CSV and PDF reports, admin forecast

Built with

Python · Streamlit · pandas · scikit-learn · joblib · SQLite · ReportLab · Open-Meteo

Secure access: login and registration





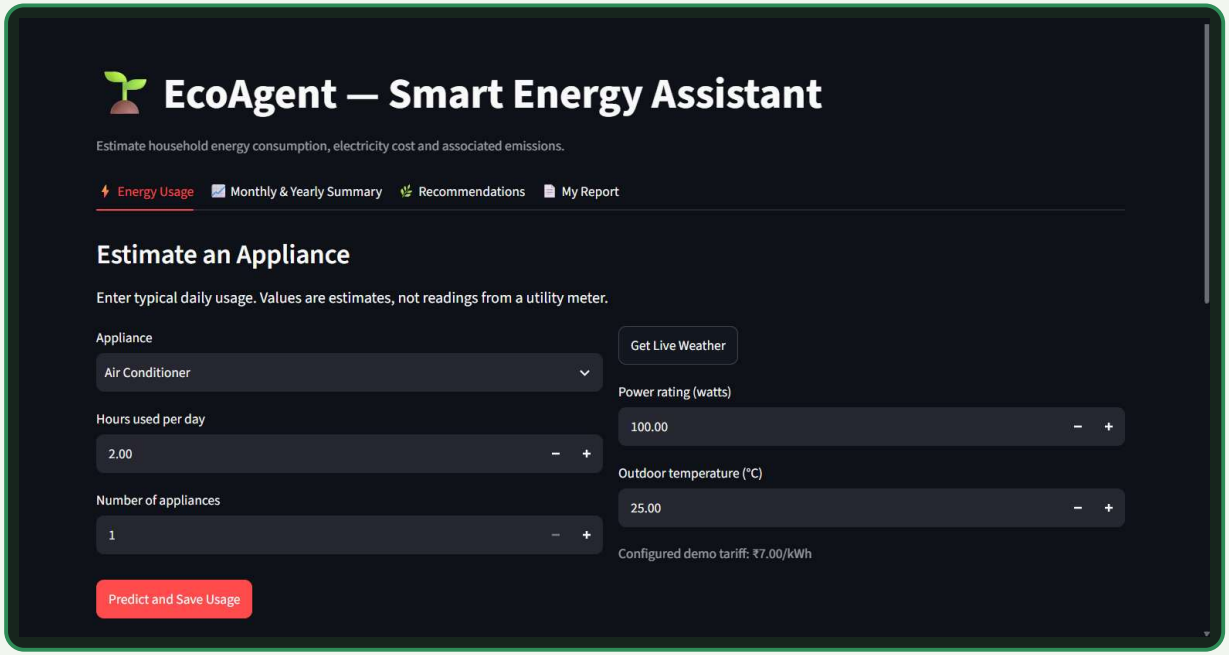
Login

- Username and password form
- Passwords are checked as SHA-256 hashes
- Users go to the user dashboard; admins use the Admin tab

Create account

- Choose a username and a password of 6+ characters
- Select a state or UT, which sets the electricity tariff
- Duplicate usernames are rejected

Energy Usage: estimate an appliance



Estimate = 0.5 × Random Forest prediction + 0.5 × (hours × quantity × watts ÷ 1000), then cost = kWh × tariff.

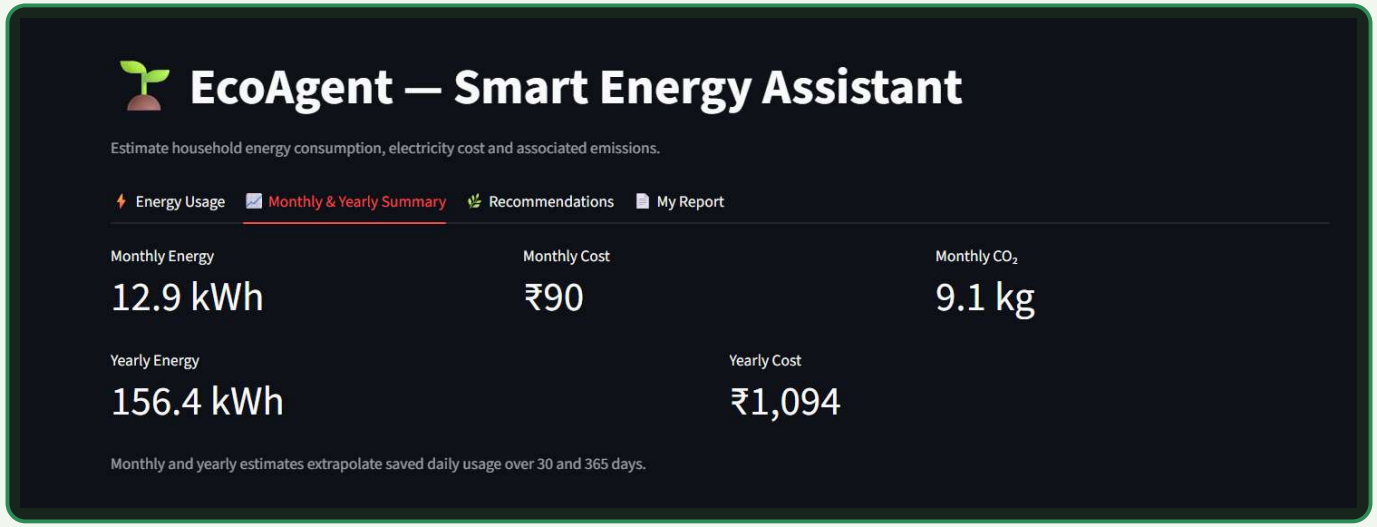
How it works

- Pick the appliance, hours per day and quantity
- Enter power rating in watts and outdoor temperature
- Get Live Weather fills temperature from Open-Meteo
- Tariff for the user's state is shown (₹7.00/kWh here)
- Predict and Save Usage stores the result

Monthly and yearly summary

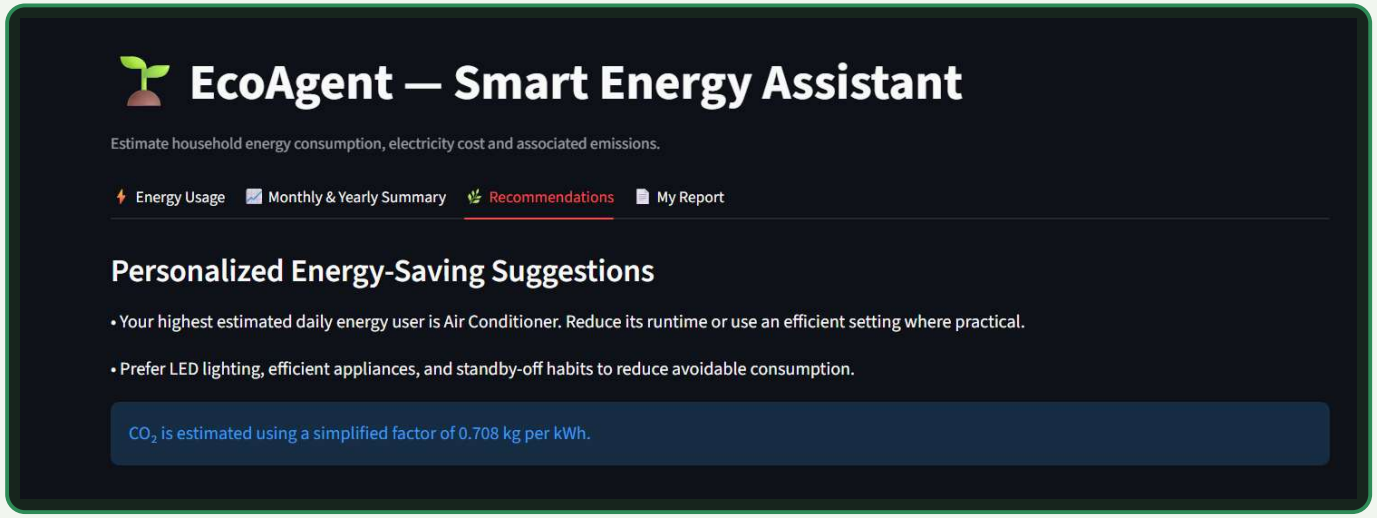
Reading the metrics

- Monthly: 12.9 kWh, ₹90, 9.1 kg CO_2
- Yearly: 156.4 kWh, ₹1,094
- Based on two saved records of 0.2142 kWh/day each
- Daily total × 30 for monthly, × 365 for yearly



CO_2 = monthly kWh × 0.708, so 12.85 kWh gives about 9.1 kg. These are extrapolations of the saved daily usage.

Personalised energy-saving suggestions



Here the Air Conditioner is the top energy user, so the first tip targets its runtime.

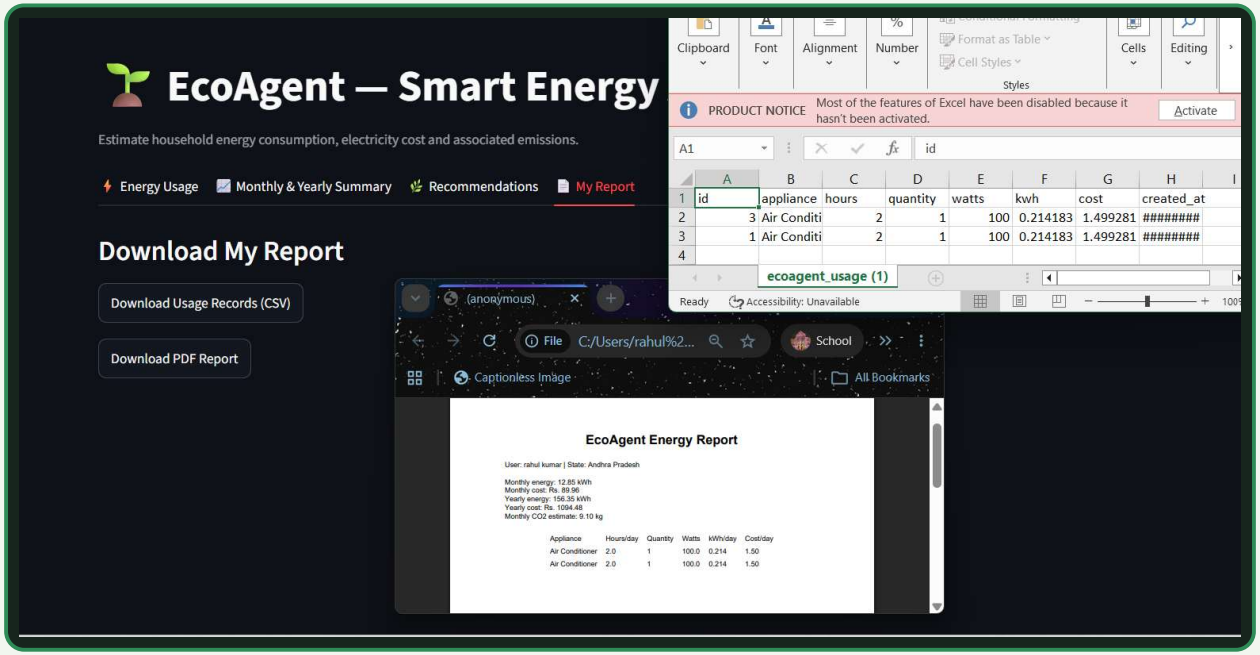
Rule-based tips

- Finds the appliance with the highest daily kWh
- Flags appliances used over 6 hours a day
- Adds general advice: LED lighting, standby-off habits
- States the CO_2 factor used (0.708 kg per kWh)

My Report: CSV and PDF downloads

Export options

- Usage records as CSV, opened here in Excel
- PDF report with user, state, summary and records table
- Built in memory with ReportLab
- Shows monthly CO_2 estimate of 9.10 kg



The PDF shows monthly energy 12.85 kWh, monthly cost Rs. 89.96, yearly energy 156.35 kWh and yearly cost Rs. 1094.48.

How a prediction is made

Example from the screenshots: Air Conditioner, 2 hours/day, 1 unit, 100 W, Andhra Pradesh

Physics formula

2 × 1 × 100 ÷ 1000

0.200 kWh

Random Forest

150 trees; inputs: hours, qty, watts, temperature

ML estimate

50/50 blend

0.5 × ML + 0.5 × physics

0.2142 kWh/day

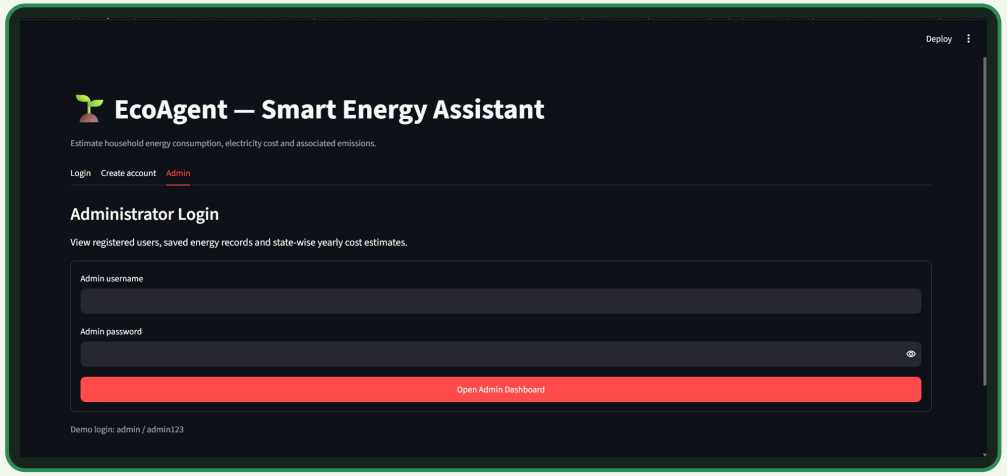
Daily cost

0.2142 × ₹7.00 tariff

₹1.4993/day

Why blend? The formula is exact and explainable; the model adds real-world effects such as temperature. The model is trained on a smart-home energy dataset (80/20 split) and checked with MAE, RMSE and R^2.

Admin: login and dashboard





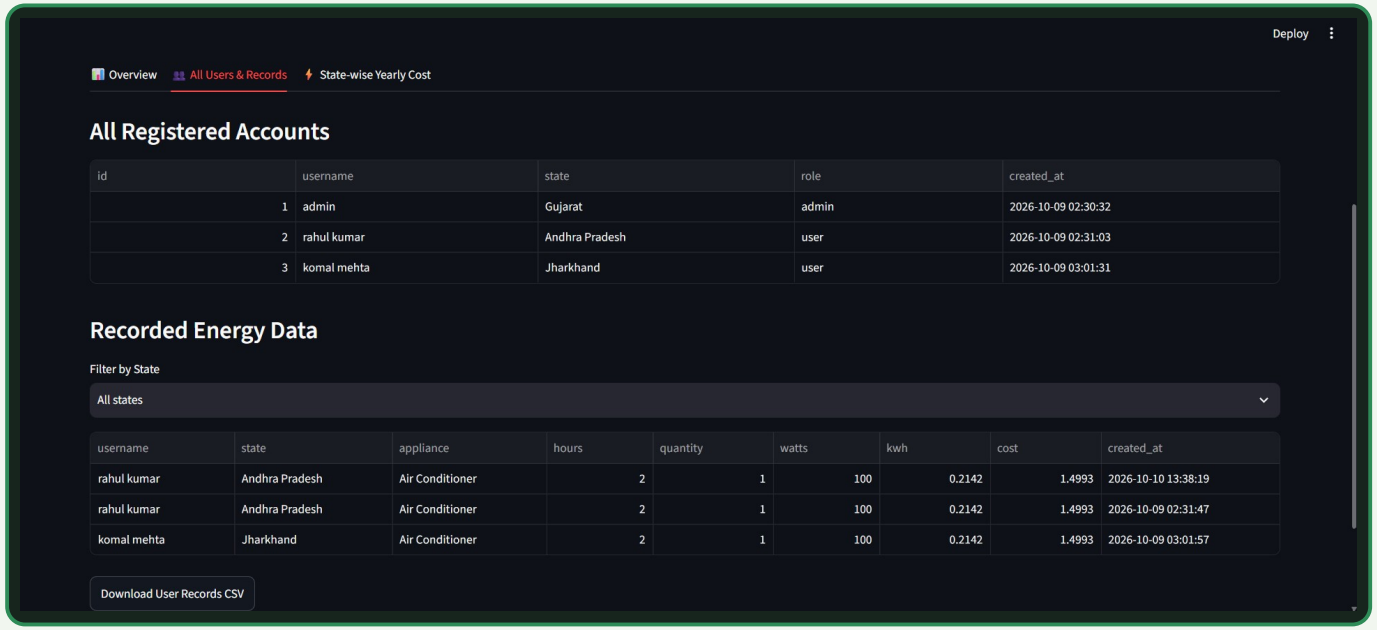
Admin login

- Separate Admin tab with its own form
- Only accounts with the admin role get in
- Demo login is shown for testing

Dashboard metrics

- 2 registered users from 2 states
- Estimated monthly energy 19.3 kWh
- Estimated yearly cost ₹1,642

All users and recorded energy data



Each record: Air Conditioner, 2 h, 1 unit, 100 W gives 0.2142 kWh and ₹1.4993 for the day.

Admin controls

- Table of all registered accounts with role and creation time
- Every saved appliance record with kWh and cost
- Filter records by state
- Download the records as CSV

Overview: users and energy per user

What the admin sees

- Registered users grouped by state, with a bar chart
- Energy summary of every user
- rahul kumar: 156.4 kWh and ₹1,094 a year
- komal mehta: 78.2 kWh and ₹547 a year



Energy Summary of Every User

username	state	monthly_kwh	monthly_cost	yearly_kwh	yearly_cost	co2_kg
rahul kumar	Andhra Pradesh	12.851	89.9569	156.3536	1094.4751	9.0985
komal mehta	Jharkhand	6.4255	44.9784	78.1768	547.2376	4.5492

Monthly and yearly kWh, cost and CO_2 for each user come from the saved summary table.

State-wise estimated yearly electricity cost



This is an estimate from registered users' records, not the real electricity demand of a state.

Forecast logic

- Groups users by state and sums yearly kWh
- Multiplies by the state tariff (default ₹7/kWh)
- Andhra Pradesh: ₹1,094.48, Jharkhand: ₹547.24
- Download the forecast as CSV

source link of project and all files

PROJECT LINK :-

https://aicteintershipgit-eqwhomuaqwsjgbtpntdvzr.streamlit.app/

FILE LINK :-

https://drive.google.com/drive/folders/1IsVW49sDpMdQA7bg5BvpMnFKY9rW9o5y?usp=sharing

Conclusion and next steps

What we built

- Appliance-wise kWh, cost and CO_2
- Hybrid ML + physics estimate
- User and admin dashboards
- CSV and PDF reports

Limitations

- Estimates, not meter readings
- Flat tariffs, no slabs or taxes
- One temperature value per record
- Simple password hashing

Future scope

- Slab-wise live tariffs
- Smart-meter or IoT data
- Seasonal forecasting
- Salted hashing, e-mail verification

Thank you